

DEVARA HEPSIBA

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BATCH-51

Lab 10 – Code Review and Quality: Using AI to Improve Code Quality and Readability

Task -1(Error Detection and Correction)

```
AI_10.PY > ...
1  def calculate_total(nums)
2  sum = 0
3  for n in nums
4  sum += n
5  return total
```

Prompt: Find syntax errors and logical errors, then fix them

Correct code:

```
#Find syntax errors and logical errors, then fix them
def calculate_total(nums):
    total = 0
    for n in nums:
        total += n
    return total
print(calculate_total([1, 2, 3, 4]))
```

Output:

```
PS C:\Users\hepsi\OneDrive\Desktop\AI>
10
```

Explanation:

- Added missing colons to fix syntax errors
- Used proper indentation
- Replaced sum with total to match the return statement
- Code now executes correctly and returns the sum of numbers

Task -2(Code Style Standardization)

Sample code:

```
def findSum(a,b):return a+b  
print(findSum(5,10))
```

Prompt: Make code clean, readable, and follow Python standards.

Correct code:

```
#Make code clean, readable, and follow Python standards.  
def find_sum(a, b):  
    return a + b  
print(find_sum(5, 10))
```

Output:

```
3/python.exe c:/Users/hepsi  
15
```

Explanation:

Instead of a single line, which improves readability and makes debugging easier.

Task-3(Code Clarity Improvement)

Sample code:

```
def f(x,y):  
    return x-y*2  
print(f(10,3))
```

Prompt:

Improve readability without changing output.

Correct code:

```
#Improve readability without changing output.
def f(x, y):
    return x - y * 2
print(f(10, 3))
```

Output:

```
PS C:\Users\hepsi\OneDrive\Desktop> python 3/python.exe c:/Users/hepsi/OneDr
4
```

Explanation:

Renamed function and variables meaningfully

Fixed indentation

Logic remains unchanged

Code is now easy to understand

Task -4(Structural Refactoring)

Sample code:

```
print("Hello Ram")
print("Hello Sita")
print("Hello Ravi")
```

Prompt: Remove repetition using reusable functions.

Correct code:

```
#Remove repetition using reusable functions.
def greet(name):
    print(f"Hello {name}")
greet("Ram")
greet("Sita")
greet("Ravi")
```

Output:

```
3/python.exe c:/Users/hepsi/O
Hello Ram
Hello Sita
Hello Ravi
```

Explanation:

Created a reusable function greet()

Reduced repetition, Easy to add more names later

Task 5: Efficiency Enhancement

Sample code:

```
numbers = []  
for i in range(1, 500000):  
    numbers.append(i * i)  
  
print(len(numbers))
```

Prompt: Optimize code for better performance.

Correct code:

```
#Optimize code for better performance.  
numbers = [i * i for i in range(1, 500000)]  
print(len(numbers))
```

Output:

```
3/python.exe c:/Users/hepsi/OneDri  
499999
```

Explanation:

Used list comprehension

Faster execution

Less memory overhead

Same output as original code

